MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular EXAMINATION

THIRD SEMESTER

MG3SDCREM200- Material Science For

Renewable Energy Technology

(2025 ADMISSION ONWARDS)

Duration: 2 hr **Maximum Marks:70**

Rememder(K) , *Understand(U), Apply(A), Analyse(An),* Evaluate(E), Create(C), Skill(S), *Interest(I) and Appreciation(Ap)*

PART A

Short answer

Answer any 25 questions

Each question carries 1 marks

1. Mention any two important mechanical properties of materials.[K, CO1]
2. State the significance of optical properties in materials. [U, CO1]
3. List any two magnetic materials. [K, CO1]
4. Identify the class of materials that exhibits electrical conductivity intermediate between conductors and insulators. [U, CO1]
5. Give a brief definition of nanomaterials. [U, CO1]
6. State the phenomenon of superconductivity. [U, CO1]
7. Mention one application of nanomaterials in solar cell technology. [A, CO1]
8. Identify the role of a nano catalyst in a chemical reaction. [U, CO1]
9. Express stress in terms of force and area. [K, CO2]
10. Express strain in terms of change in dimension and original dimension. [K, CO2]
11. State Hooke's law. [K, CO2]
12. Distinguish tensile stress from compressive stress. [U, CO2]
13. Mention the condition under which thermal stress develops in a material. [U, CO2]
14. State the property by virtue of which a material regains its original shape after removal of the applied load. [K, CO2]

15. Identify the type of deformation that remains permanent after removal of the applied load. [K, CO2]

16. Indicate the significance of yield stress in material behaviour. [U, CO2]

17. Mention the effect of strain hardening on the strength of a material. [U, CO2]

18. List any two environmental factors that affect materials. [K, CO3]

19. Identify the electrochemical deterioration of a metal due to its interaction with the environment. [K, CO3]

20. Mention the two broad types of fracture based on the mode of failure. [K, CO3]

21. State the purpose of failure analysis of solar materials. [U, CO3]

22. Classify polymers based on their source. [U, CO4]

23. List any two examples of synthetic polymers.[K, CO4]

24. Identify the class of polymers that can be repeatedly softened by heating and hardened by cooling. [U, CO4]

25. Mention any two polymers commonly used in engineering applications. [K, CO4]

26. State the purpose of vulcanization of rubber. [U, CO4]

27. Mention one environmental advantage of biodegradable polymers. [E, CO5]

PART B

Short Essay

Answer any 5 questions

Each question carries 5 marks

28. Discuss the mechanical, thermal and electrical properties of metals with suitable examples. [U, CO1]

29. Describe the phenomenon of superconductivity and outline its major applications. [U, CO1]

30. Discuss the role of nanomaterials in renewable energy applications, with special reference to solar cells, sensors and hydrogen generation. [A, CO1]

31. Explain the concepts of stress and strain and derive the relationship between them based on Hooke’s law. Illustrate the stress–strain behaviour of a material. [An, CO2]

32. Discuss elasticity, plastic deformation and yield stress in metals and polymers. [An, CO2]

33. Examine the major environmental effects on materials, with particular reference to corrosion, erosion, thermal stress and weathering. [An, CO3]

34. Classify polymers into natural and synthetic types and compare their characteristics with suitable examples. [E, CO4]

PART C

Essay

Answer any two questions

Each question carries 10 marks

35. Discuss the mechanical, optical, thermal, electrical and magnetic properties of materials, highlighting their significance in engineering applications. [U, CO1]

36. Describe the stress–strain diagram of a material. Discuss thermal stress, elasticity, plastic deformation, yield stress, shear strength and strengthening mechanisms. [An, CO2]

37. Analyse the effect of temperature on material behaviour. Discuss fracture behaviour of different materials and the significance of failure analysis in solar materials. [E, CO3]

38. Describe the structure and properties of natural rubber. Explain the process of vulcanization and discuss synthetic rubbers such as SBR and neoprene. Add a note on the environmental impact of plastics and biodegradable polymers. [E, CO5]